Roll No. | | | | | | | |

END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering
Name of the Paper: Heat and Mass Transfer

Semester: IV
Course Code: TPE-403

Time: Three Hours

MM: 100

Note:

v. All questions are compulsory.
vi. Answer any two sub questions in each main question.
vii. Total marks for each main question is **twenty**
viii. Each Question carries **10 marks**

Q1		COs
a.	What do you mean by extended surface for the rate of heat transfer (fins)? What are the types of fins (diagram needed)? Explain the uses of fins?	CO1, CO2
b.	Illustrate the logarithmic mean area for the 'plane wall and cylindrical shape' and 'cylindrical shape and spherical shape' where the rate of heat transfer taking place.	CO1, CO2
c.	A cylindrical tank with hemispherical ends is used to store liquid oxygen at -183 °C. The diameter of the tank is 1.5 m and the total length is 8 m. The tank is covered with a 10 cm thick layer of insulation. Determine the thermal conductivity of the insulation, so that the boil-off rate does not exceed 10.8 kg/h. The latent heat of vaporization of liquid oxygen is 217 kJ/kg. Assume that the outer surface temperature of the insulation is 27 °C and that the thermal resistance of the wall of the tank is negligible.	CO3
Q2		
a.	The wall of a cold storage consists of three layers –an outer layer of ordinary bricks, 25cm thick, a middle layer of cork, 10cm thick and an inner layer of cement, 6cm thick. The thermal conductivities of materials are –brick : 0.7, cork: 0.043, and cement: 0.72 W/m^{0}C. The temperature of the outer surface wall is 30^{0}C and that of the inner is -15^{0}C.Calculate (i) the steady state rate of heat gain per unit area of the wall, (ii) the temperature at the interfaces of the composite wall, and (iii) the percentage of the total heat transfer resistance offered by the individual layers. What additional thickness of cork should be provided to make the rate of heat transfer 30 percent less than the present value?	CO3, CO4
b.	Discuss in details for the qualitative descriptions of free convection flows? Explain the different free convection boundary layer with the help of neat and clean diagram.	CO1, CO4
c.	Prove that the net heat exchange for the radiation heat transfer rate (Q)	CO1,

	between the two walls with 'n' shields is equal to the [1/(n+1)]Q without shield. Specify the symbol used in deriving the relations.	CO2
Q3		
a.	Determine the rate of heat loss by radiation from a steel tube of outside diameter 70 mm and 3 m long at temperature of 227 °C if the tube is located within a square brick conduits of 0.3 m side and at 27 °C. Take emissivity for steel and brick as 0.79 and 0.93 respectively.	CO1
b.	Solve the following: a. A refractory material which has $\varepsilon = 0.4$ at 1500 K and $\varepsilon = 0.43$ at 1420 K is exposed to black furnace walls at 1500 K. What is the rate of gain of radiation per m^2 area? b. Define radiation Shield? Determine the radiant heat exchanger in W/m^2 between two large parallel steel plates of emissivities 0.8 and 0.5 held at a temperature of 1000 K and 500 K respectively, if a thin copper plate of emissivity 0.1 introduced as a radiation shield between the two plates. Use $\sigma = 5.67 \times 10^{-8}$ $W/m^2 K^4$.	CO1, CO2
c.	Derive an expression for the heat exchange between the non black bodies for infinite long concentric cylinder Specify the symbol used.	CO2, CO5
Q4		
a.	What are the types of Heat exchanger on the basis of nature of (i) heat exchange process, relative (ii) relative direction of fluid motion and (iii) Design and constructional features?	CO1, CO5
b.	A counter flow double pipe heat exchanger using superheated steam is used to hot water at the rate of 10500 kg/h. The steam enters the heat exchanger at 180. ^{0}C and leaves at 130 ^{0}C. The inlet and exit temperature of water are 30 ^{0}C and 80 ^{0}C respectively. If the overall hear transfer coefficient from steam to water is 814 W/m. ^{0}C, calculate the heat transfer area. What would be the increase in the area if the fluid flows were parallel?	CO4, CO5
c.	Derive an expression for the effectiveness of a parallel flow heat exchanger in terms of the number of transfer units, NTU, and the capacity ratio C_{min}/C_{max}.	CO3, CO6
Q5		
a.	Discuss steady state diffusion through a plane membrane and hence find the mass transfer rate expressions for the same.	CO4, CO6
b.	Derive an expression for the mass flow rate the steady state diffusion through a cylindrical shell	CO1, CO5
c.	Explain in details about steady state equimolar counter diffusion? Hence derive the expression for the same.	CO3, CO4